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MétaCan: study protocol

An open, provenance-tracked, coverage-audited map of Canadian metaresearch

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Status: preregistration draft. To be registered on OSF **before the frame is built and before any retrieval is run**. Registration date, OpenAlex snapshot version, and Érudit harvest date will be stamped into `RELEASE.md` at v1.0 and cannot be revised afterwards.

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1. Rationale

Mapping a research community with bibliographic metadata invites a specific failure: the map records what the retrieval could see, then quietly presents itself as a map of the field. The gap between the two is rarely measured, and where it is measured it is often large. In earlier work I benchmarked automated publication linkage for funded trials against a manual reference standard built by two independent reviewers. Conditional on a candidate reaching the matcher, it performed well (sensitivity 92.2%, PPV 94.3%); unconditionally, its sensitivity was **44.5%**. The failure was in retrieval, and it was invisible to every internal quality check.

MétaCan is built so that the same failure, if it occurs, is measured rather than concealed. Two commitments follow, and they drive every design decision below:

1. **Provenance.** Each record states which route(s) retrieved it and which sense of "Canadian" it satisfies.
2. **A measured gap.** A human audit samples the records the pipeline did *not* retrieve, so that retrieval sensitivity is estimated rather than assumed.

2. Preliminary work (complete, and reported in the proposal)

The pilot analyses are reproducible via `make pilot` and re-derivable offline from archived responses via `make pilot-offline`. The authoritative list, with every value, is **`pilot/results/FINDINGS.md`**, which is *generated* from `pilot/results/findings.json`; neither the list nor its COUNT is restated here, because a hand-maintained copy of a generated table is a stale table waiting to happen. This protocol previously carried a seven-row version of it, which was already wrong by the time anyone read it, and then a hand-typed count, which went stale one sentence away from this warning.

The five that determine the design:

#	Result	Design consequence
12	Scored against the rubric, the best topic route retrieves **12%** of Canadian metaresearch (95% CI 5.6 to 21.6), at 60% precision. It misses **66 of 75**. OpenAlex files a work by what it is *about*, so metaresearch about cardiology reads as cardiology.	**The field is invisible to topic retrieval precisely because it is about other fields.** Retrieval is demoted to provenance; screening over a Canadian frame is the gate.

#	Result	Design consequence
10	Swap which model is "the screener" and the base rate moves from **1.06% to 2.37%** (design-weighted, same 1,290 records): 37,032 vs 83,022 works. The published binomial CI contains neither.	Machine agreement is a **process metric**, not accuracy. The screener-swap range, not the binomial interval, is the honest uncertainty. The human audit is the study.
11	**31.5%** of the pilot partition (**23.3%** of the built frame: 1,003,117 works, finding 19) has no abstract, and the screen finds 0.78% metaresearch there against 1.55% where one exists ($p = 0.023$; robust to adjustment for year and language). An earlier version of this table also claimed the blindness was **differential** (T2 losing 3.6x against T1's 1.4x). **That cell holds four works and the interaction is not significant ($p = 0.141$). WITHDRAWN** (DEVIATIONS.md D6).	A third of the frame is screened on its title alone and the screen finds half as much there. The audit stratifies on abstract availability. It is **not** evidence of differential blindness by tradition, and this protocol no longer says it is.
14	A simple random sample of the screened-out mass **cannot measure screening sensitivity**: 600 records return an expected **0.4** misses; twenty would need **2,009 coder-hours** against 65 budgeted.	The two-phase audit as first specified was impossible. **See §6.0 and §6.1.**
16	Haiku lands near Sonnet's base rate (1.27% vs 1.06%, 98.1% agreement) but their in-scope sets overlap **16%** unweighted, **10%** design-weighted; of Sonnet's 58 positives Haiku confirms **12**. And agents of ONE model on ONE prompt disagree beyond chance after conditioning on what they were shown (CMH $p =$ **0.0056**, replicated at **0.015**), with the agents' ordering flipping between arms.	**Rate agreement is not set agreement, and no machine pass measures this field.** The machine screen is demoted to a **stratifier** for the audit: design weights stay unbiased under a noisy stratifier, only efficiency suffers, and power is computed at Haiku-realistic concentration. Machine tiers are released marked provisional; every reported prevalence is design-weighted human coding.

#	Result	Design consequence
D1	**The pilot screen did not obey this rubric.** It was sent 6 of the 8 fields §5 mandates: **venue, OpenAlex topic and field, Canadian affiliations and funders were all withheld.** Every pilot number came from a title-and-abstract screen.	The most serious defect in the project. The full screen sends the rubric's complete payload and reports the difference as a finding. **See DEVIATIONS.md D1.**

Also established: 0 of 4,516 OpenAlex topics name the field; 64% of works in the topic space have no raw affiliation string; Érudit matches 0 OpenAlex sources but exposes 379 harvestable OAI-PMH sets; `reproducibility` alone returns 43,392 Canadian works at 0.8% on-topic; NSERC carries 9.2x SSHRC's linked works; naive capture-recapture implies Canada produces 59% of world metaresearch against an observed 1.9%, and is **abandoned**; the OpenAlex API is now metered, so the frame is read from a pinned S3 snapshot; and screening the frame with the full rubric twice costs **~$4,767** on the Batch API (§7). An earlier version of that estimate charged the rubric once per WORK rather than once per CALL, reported $24,379, and used it to justify a cheap TRIAGE in front of the screen. The triage is a retrieval step, in a study whose central finding is that retrieval destroys these maps. With the arithmetic corrected it saves $110, so **it is deleted**: every work in the frame is read against the full rubric. (DEVIATIONS.md D7.)

3. Estimand (prespecified)

The field boundary is contestable. That is a reason to decide in advance and be explicit, not a reason to defer.

Primary estimand: Canadian-*produced* metaresearch. A work is in scope if it is classified T1 or T2 (§5.2) **and** satisfies at least one of: - **CA-AFF**: ≥1 authorship with an institutional affiliation whose country is Canada, **or** whose OpenAlex-parsed authorship country is Canada even where the institution is unresolved (the second clause exists because finding 2 shows 64% of the topic space carries no resolvable affiliation); - **CA-FUND**: ≥1 funder whose OpenAlex `country_code` is Canada. This is an **external criterion, not a curated list**: an earlier version of this clause named four funders (CIHR, NSERC, SSHRC, CFI), and that list failed twice: three of its four hardcoded IDs pointed at *other funders*, and even corrected it missed Canada Research Chairs (42,824 works), the Government of Canada, NRC, Mitacs, Genome Canada, and every provincial agency. OpenAlex knows **2,182** Canadian funders; `R/pin_entities.R` pins them, and the pinned list ships with the data.

The two components are **reported separately as well as jointly**, because pilot finding 6 shows CA-FUND is skewed against the social sciences, and a pooled figure would hide that.

Secondary estimand: metaresearch *about* the Canadian research system, irrespective of author location, identified by screening rather than by keyword.

Descriptive attributes, NOT inclusion criteria. Recorded on every record, never used to admit one: - **CA-VENUE**: published in a Canadian venue (incl. Érudit-hosted); - **CA-PERSON**: an author with a known prior Canadian affiliation, at any time.

Rationale for the exclusion: a work by someone who once held a Canadian post, written abroad, with no Canadian funding, affiliation, subject matter, or venue, is not self-evidently Canadian metaresearch.

Nor is a globally generic paper that happens to appear in a Canadian journal. These are interesting covariates and poor gatekeepers.

Prespecified alternative rules (to test the consequences of the primary decision, not to replace it): - **ALT-1 (strict):** CA-AFF only. - **ALT-2 (broad):** primary ∪ CA-VENUE ∪ CA-PERSON.

Only these two alternatives will be reported. We will not enumerate all combinations of the five flags: that is researcher degrees of freedom wearing the costume of a sensitivity analysis.

4. Frame (F)

F is defined and frozen before screening, and every claim in this study is bounded to F.

The design inverts the usual one. The obvious approach retrieves what *looks like* metaresearch and then asks whether it is Canadian. That makes the field boundary a property of the retrieval strategy, and it is why such maps cannot be audited: a lexicon cannot show you what it never surfaced, so there is nothing left to measure the miss against.

Here the frame is **Canadian research as four checkable metadata routes see it** (affiliation, funder country, venue, aboutness): an external, enumerable criterion that owes nothing to our notion of metaresearch. It is not literally "all Canadian research": it admits about-Canada works with no Canadian production, and it cannot see works invisible to all four routes; both facts are measured rather than wished away (findings 7 and 27). Field membership then becomes a **classification** question over an enumerable universe, not a **retrieval** question over the literature. That is what makes "what did we miss?" answerable at all, and it is what produced the base rate (§2, finding 9) that replaces the failed coverage estimator.

F IS NO LONGER A HYPOTHESIS. IT IS BUILT. All 482 partitions of the pinned snapshot were streamed and filtered: **F holds 4,299,418 works**, each appearing exactly once (finding 23). For most of this project's life F was "about 3.5M", an **extrapolation from a single partition** that sat hardcoded in six scripts; **the estimate was 18% low**, and every cost, audit-power and field-size figure computed against it has moved. `R/frame_size.R` now reads the size from the artifact the harvest writes, so no script can hardcode it again.

Sources that CONTRIBUTE RECORDS to F. - A **pinned OpenAlex snapshot** (release stamped; not the live API, which is metered and mutable. See finding 8: 1,000 credits per ~11h window, and 8.2 days per pass. A study behind a meter that resets every eleven hours cannot be re-run by a reviewer). - A **pinned Érudit OAI-PMH harvest** (`https://oai.erudit.org/oai/`, 379 sets; harvest date stamped).

Sources that DO NOT contribute records, and must not. Retraction Watch, ClinicalTrials.gov, and the CIHR project database are used, and **none of them admits a work to F**. A retracted cardiology paper is retracted *cardiology*; a trial registration is not a publication; a grant is not a work. Admitting them would let an interesting signal masquerade as the estimand, which is the error this protocol exists to prevent. They enter as: - **Record attributes.** Retraction Watch supplies the post-publication state (finding 17): OpenAlex's `is_retracted` is a *boolean over a four-value space* (retraction, expression of concern, correction, reinstatement), so it expresses one and silently reports the rest as false. RW is joined by DOI and its state and reasons ship with F. - **Reference standards.** ClinicalTrials.gov knows a Canadian trial happened *independently of any pipeline*, so it cannot be wrong in the pipeline's favour. It is the only reference standard here **not made of machine labels** (finding 21), and §6.1 uses it for known-item recall alongside the venue set. - **Enrichment.** PubMed / Europe PMC / Crossref supply abstracts OpenAlex lacks (finding 19). **Preprints are already in F** and need no ingest: measured against bioRxiv and medRxiv's own API, OpenAlex indexes 99.6% of what the servers hold (finding 20).

Window. Publication years 2000–2025 inclusive.

Membership. A record enters F if it carries **any identifiable Canadian signal**: CA-AFF · CA-FUND · Canada named in title, abstract or keywords (EN/FR) · published in a Canadian venue · is an Érudit record. Every record stores **which routes admitted it**. **1,565,226 works (36.4% of F) carry no Canadian affiliation at all**: an affiliation-only frame would hold 2,734,192 works and would never see them. That is the empirical case for the union of routes, and for provenance on every record.

Limitations, stated up front. - A work whose Canadian link is invisible to the metadata (no affiliation, no funder, no textual mention, no Canadian venue) cannot enter *any* Canadian frame, by any method. That is a hard boundary of the data, not of this design. - Scholarship indexed by neither OpenAlex nor Érudit is **not estimated** here, and no claim about it will be made. - **Both clauses of the primary estimand rest on sparse metadata.** 64% of the metaresearch topic space carries no raw affiliation string (finding 2), and **71.2% of F carries no funder metadata at all** (finding 18). CA-FUND cannot admit a work whose funder OpenAlex never recorded; that is a hard ceiling on the route, and it is why the audit must sample the works **no route reached**.

5. Screening (retrieval is demoted to provenance)

Every record in F is **classified**, not retrieved. Retrieval routes still run (candidate topics and venues; bilingual lexical and semantic similarity to the seed corpus; citation neighbours of curated seeds), and every record stores the set of routes that surfaced it. But those routes are **provenance, not the gate**: they answer "how would a conventional search have found this?", which is exactly the question the coverage audit needs, and they are never allowed to decide membership.

Route overlap and incremental yield are reported descriptively. They are **not** used to estimate the unseen population (§6.3).

5.2 Inclusion tiers

- **T1 (core):** the work's object of study is research itself: its methods, reporting, reproducibility, integrity, evaluation, funding, workforce, communication, or infrastructure.
- **T2 (adjacent):** empirically or conceptually grounded work on science as a social system (STS, library and information science, research infrastructure studies) that is not framed as metaresearch but studies it.
- **T3 (contextual):** policy documents, editorials, commentary, announcements. **Mapped and released, never pooled into the analytic corpus.**
- **Excluded:** everything else.

5.3 Genre label

`empirical` · `conceptual` · `editorial/commentary` · `policy` · `infrastructure/announcement` · `other`. Assigned by the classifier because OpenAlex's `type` is unreliable and this field is unusually commentary-heavy.

5.4 Classifier

Multilingual sentence embeddings (cross-lingual by construction) over title + abstract, plus an LLM screening pass with **published prompts** and temperature 0. The decision **threshold is locked before evaluation** and recorded in the protocol registration.

The classifier plays no part in constructing the reference standard against which it is judged (§6.2). No LLM output is shown to the human coders.

6. Validation: two-phase stratified probability audit

The core of the study, and the thing the field usually skips.

5.5 The classifier: what it may do, what it may not, and what it is actually for

It was proposed as a way to avoid LLM-labelling 4.3M works, and that premise is false. Finding 13 measured the cost: the full v3.1 rubric over **every work in the frame is $1,261**. There was never anything to avoid, and a proposal that budgeted the full screen in one paragraph while justifying a cheap substitute for it in the next was describing two studies (D32). The screen reads every work. The classifier is a **different instrument**, and it is preregistered here as one.

What it may not do: emit a category label. Not thresholded, not "provisional", not for the confident core. Distilled from our own metaresearch labels, a student reproduced **its own teacher's positive set at Jaccard 0.17** (finding 25). Two independent adversarial reviews reached the same verdict by different routes: per-teacher heads are useful for **allocating human effort** and **cosmetic as a solution to the contested boundary**. Scores ship; labels do not.

What it is for, each measured rather than asserted (finding 30):

1. **A calibrated score for the audit to stratify on.** Top-decile lift 3.9x to 5.3x. Tier labels are discrete and uncalibrated; the score-strata need a continuous quantity.
2. **Rubric revisions testable in minutes** instead of one $1,261 pass each.
3. **A frame-wide estimate of where the three teachers would split**, from 5,600 labels rather than three full passes (3x cost, ~25 days). It works and **it is weak: 3.74x the base rate**, reported as an efficiency gain for the disagreement stratum, never as a map to trust on its own. Shared teacher bias is invisible to it: if all three are wrong the same way, the spread is zero and the work looks settled.
4. **Learnability as evidence about the boundary**, which produced the finding below.

Three results that constrain everything downstream.

- **The estimand moves measured performance 2.7x** (design-weighted AP: unanimous 0.078, majority 0.136, any 0.209). There is no "the" target. Every consensus rule is a **named sensitivity analysis**, and training on one silently settles the question the audit exists to answer.
- **The unanimous core is the HARDEST to learn, not the easiest.** If the boundary were a line the models share with noise around it, the agreed core would be the separable part. It is not.
- **The abstract is worth more than any modelling choice** (+0.115 AP, 84% relative), and the frame cannot serve it: the works table stores `has_abstract`, a boolean, and no text. The **deployable model is the worse one**, `ml/features.py` **FAILS the build** if a model is trained on a field inference cannot supply, and the gap is now a priced budget decision rather than an assumption.

Active learning (finding 31). Batches of 100 = 50 max-teacher-disagreement + 30 max-uncertainty + 20 random. Against a **random-batch control at identical budget**, active acquisition takes held-out AP from 0.014 to **0.139** where the control reaches 0.050: **2.8x, winning 18 of 20 rounds**, with held-out churn falling to 0.11. **The 20 random anchors are NOT an evaluation stream**: after 400 draws they hold **three** positives, and the 95% interval on prevalence is **wider than the prevalence**. They are demoted to **drift sentinels**, never used to select a batch. **The loop learns; the audit measures.** AP throughout is agreement with the majority *teacher*, so the entire curve is **imitation, not accuracy**, and the simulation runs on labels that already exist: it shows the loop's mechanics and **cannot** show that it matures on 4.3M unlabelled works.

Study design, the one annotation with an external reference standard. MEDLINE publication types are assigned by NLM independently of this project, so a design label can be **wrong in a way the pipeline cannot hide**. Three constraints, all prespecified:

- **The reference subset is defined by NLM's record-level `IndexingMethod`, not by a year.** Full human indexing ended as universal practice around 2011 and automation completed in 2022; the eras overlap for a decade, so no date cutoff separates them. Absent = human (the reference); `Curated` = automated-then-reviewed; `Automated` = machine. The latter two measure **agreement with another machine** and are never pooled into the gate.
- **Publication types are overlapping binary attributes, not an exhaustive mutually-exclusive ontology.** A work can be both Systematic Review and Meta-Analysis. Six **independent binary heads**, never a softmax that forces a choice the standard itself does not make.
- **Class-specific calendar eligibility, because a type that did not exist cannot be a negative.** *Observational Study* enters the vocabulary in **2014**, *Systematic Review* in **2019**. A 2008 cohort study carries no Observational Study tag because the tag did not exist. Records outside a class's window are **not data** for that class, in either direction.
- **The gate is on the lower 95% confidence bound**, not the point estimate, with a minimum eligible sample; a class that fails ships as a **score, marked unvalidated**.

6.0 The design this replaces, and why it could not work

An earlier version of this protocol said: partition F into screened-in and screened-out, draw a probability sample from both, human-code them, and estimate sensitivity with design weights. That is the textbook two-phase audit and it is **arithmetically incapable of doing the job here.** Finding 14 does the sum.

The works a screen wrongly rejects are a vanishing fraction of the rejected mass. At 95% screen recall, the ~4.24M screened-out records contain ~2,300 missed works: a density of **0.07%**. A sample of 600 screened-out records therefore returns an **expected 0.4 of them**, and you cannot estimate a rate from zero events. Seeing twenty would take **30,134** human codings, about **2,009 coder-hours** against the ~65 this study can buy.

So the central promise of the project (a human audit that quantifies what the pipeline missed) was, as first specified, impossible. It is recorded here rather than quietly fixed, because a preregistration that hides its own false starts is not a preregistration.

6.1 Design: three instruments, not one

(i) Score-stratified probability sampling of the screened-out set. This is salvageable, and only because the misses **concentrate**. The pilot measured where: of the 37 works screener B pulled into scope that screener A rejected, **30 sit in the contested boundary and only 6 in the settled rejects**. So:

- Partition **F** into **screened-in** (any v3 category) and **screened-out** (empty `categories`).
- Draw a **stratified probability sample from both**, with **known, recorded selection probabilities**. Strata cross: `language (EN/FR/other)` × `source (OpenAlex/Érudit)` × `partition` × `screener score` × `abstract availability`.
- Oversample heavily: the **contested boundary** (where the pilot shows the misses concentrate); French records; Érudit records; adjacent STS/LIS material; **records with no abstract** (finding 11); under-represented provinces and genres.
- Sample size set to achieve a target half-width on screening sensitivity within the French stratum, the scarcest and therefore binding cell. Computed and fixed at registration.

Why stratification is binding rather than decorative: at a ~1.3% base rate a uniform sample spends its whole budget confirming obvious exclusions. Without stratification the audit is uninformative at any feasible n.

What the human coders see, and what the audit can therefore claim. Blinding the coders to machine labels prevents contamination; it does not create information. Two humans independently reading the same missing abstract cannot discover metaresearch that only the full text reveals, and design weights correct selection probabilities, not outcome misclassification. So: (a) sampled records are coded on the **best ascertainable evidence**, the finding-19 full-text cascade (PubMed, Europe PMC, Crossref) plus publisher pages where the cascade fails, not on the models' own payload; (b) the coding scheme includes an **`unresolved`** outcome, and unresolved is reported, never imputed; (c) every estimate is stated **conditional on ascertainment**, with the unresolved fraction alongside it; (d) the release is layered: **human-verified / high-precision / provisional / contested**, so no layer borrows another's authority. The audit estimates agreement between humans applying this rubric to the evidence they could obtain; that sentence, not "true sensitivity", is what the numbers mean, and the protocol says so before the numbers exist.

(ii) Recall against already-labelled data, not against the haystack. Any filter's recall (the topic route, the lexical route, the cheap triage in finding 13) is measured directly against the **5,737 works that already carry full-rubric labels**: run the filter over them and count what it drops. This is exactly how finding 12 scored the topic route at 12%. It costs nothing and needs no needle-hunting in the discarded mass. **No filter enters the pipeline without its recall measured this way first.**

(iii) Known-item recall on an external criterion, because (i) is blind exactly where the bias is.

Score-stratified sampling finds the works the screener *almost* caught. It is structurally blind to the ones it rejected *confidently*, and finding 11 identifies precisely which those are: T2 material (STS, LIS, the humanities) with no abstract, rejected on a title carrying none of the field's vocabulary. Such a work is not near the threshold. It is deep in the settled rejects, where (i) never looks. **An instrument that is blind where the bias lives is worse than no instrument, because it returns a confident number.**

So a third instrument, on a criterion no screener can be blind to: **venue**. A reference set of Canadian-authored articles in *Social Studies of Science*, *Scientometrics*, *Quantitative Science Studies*, *Research Integrity and Peer Review*, JASIST, *Research Evaluation*, *Accountability in Research*, *Recherches qualitatives*, *Documentation et bibliothèques*. **A Canadian-authored paper in *Social Studies of Science* is T2 by where it was published, whatever its abstract is about.**

Venue is immune to *aboutness*, which is the single thing that defeats topic retrieval (finding 12) and title-only screening (finding 11) alike. This is the frame flip applied a second time: replace "does this look like metaresearch?" with an external, checkable criterion. It is the only instrument that can see into the blind spot.

What (iii) is and is not. Known-item recall on a venue reference set is a **non-probability** estimate. It bounds and diagnoses recall on the hard cases; it does **not** replace the design-weighted population estimate from (i). Both are reported, separately, and neither is presented as the other.

6.1a A note on why agreement statistics are stratified

With a 1.3% positive rate, two screeners that both label the obvious 98% as OUT achieve high raw agreement and an unstable κ: the statistic is dominated by the cell where agreement is free. This is the well-known κ paradox with skewed marginals. Agreement is therefore reported **per stratum** and reconstructed with design weights, never read off a pooled 2×2. The pilot already shows why this matters: agreement is 99% in the settled stratum but 95% at the boundary.

6.2 Coding

- **Two HUMAN coders, independent, blinded** to the machine labels and to each other.
- A **locked rubric** (`docs/protocol/rubric.md`, versioned) with worked examples and explicit edge cases.
- Disagreements adjudicated against the rubric; adjudication decisions logged.
- Agreement reported (Cohen's κ / Krippendorff's α) **as a property of the coding, not as evidence of classifier validity**: an agreement coefficient measures coders, not the instrument.
- **Dependency, declared:** the second coder must be bilingual (FR/EN) for the French stratum. To be recruited and **compensated** in Week 1 via CaRN, ACFAS, CAIS-ACSI, CARL. **If no francophone coder is recruited, the French-stratum estimates are withdrawn, not imputed.**

6.2a What the machine screen does, and what it cannot do

One full-frame LLM pass (a cheap model) labels every work in F against the locked rubric; a second model relabels a 20,000-record stratified sample. Disagreements are recorded and routed to the human audit. Inter-model agreement is published and **reported as a process metric**.

It is not, and will not be presented as, an accuracy estimate, and after finding 16 it is not presented as a *measurement* of anything. Agreement fails twice. First, two language models share training data, lexical priors and failure modes; their errors are correlated, and most correlated precisely on the boundary cases the field's definition turns on. Second, finding 16: **rate agreement is not set agreement**. The cheap model matches Sonnet's base rate within a fraction of a point and agrees with it on 98.1% of the frame, while their in-scope sets overlap 16% (10% design-weighted); and agents of one model, on one rubric and prompt, disagree beyond chance after conditioning on what each was shown, in two arms, with their ordering flipping between arms. **This is not "duplicate screening"**: that term's warrant comes from independent *human* judgement, and borrowing it for two models would be a misrepresentation of method. The accuracy claim rests on §6.2 and nowhere else.

What the machine pass *does* contribute is real and, after finding 16, precisely bounded: **stratification** (a noisy stratifier costs the audit efficiency, never validity, because design weights use known selection probabilities), an empirically located boundary (the disagreements), and the throughput to organize 4.3M works for sampling at all. **Machine tiers are released marked provisional. Every prevalence this study reports is design-weighted human coding. No machine number is reported as the**

field. The pilot's machine base rate (§2, finding 9) stands as the motivating hypothesis the audit tests, not as a result the audit inherits.

Agent-level controls, because finding 16 and D11 demand them: agents receive randomized, fixed-size chunks recorded in an assignment manifest at run time; a duplicate-agent reliability arm re-screens a common subsample; and completeness checks verify every output file against the filesystem and fail loudly on any unreturned record, because one pilot agent reported six label files it never wrote.

6.3 Estimation

Using **design weights** (inverse selection probability): - **retrieval sensitivity within F**: the headline quantity, and the answer to "what did we miss?"; - screening **precision and recall**, with confusion-matrix counts; - **eligible count within F**, with a confidence interval; - **differential error by language, source, tradition, genre, and linkage type**: i.e. *for whom* the pipeline fails, not merely how often.

All reported with confusion-matrix counts and interval estimates, not bare percentages.

Capture–recapture is not used, and this is a considered decision, not an oversight. The routes are endogenous: R3 depends on the seed set and any author-based expansion depends on authors surfaced by R1/R2. Log-linear multi-list models cannot identify the all-zero cell without an untestable restriction on the highest-order interaction, and with endogenously constructed lists that restriction is not credible. Nor is the estimate a safe lower bound: depending on the dependence structure it may overstate the unseen population as readily as understate it. Pilot finding 7 shows the naive estimator returning a figure that implies Canada produces the majority of the world's metaresearch. The only unconditional lower bound on the population is **the number of eligible works actually observed**, and that is what we will report. The two-phase audit replaces it because it samples the gap instead of inferring it.

7. Analysis (secondary and exploratory)

Everything in this section is **descriptive, secondary, and explicitly labelled as such**. None of it defines the field; the corpus is constructed and validated first, and only then described.

- Counts and trends by year, language, province, institution, genre, tier, and linkage type.
- Co-authorship and institutional collaboration structure (igraph; Louvain communities, betweenness, assortativity).
- Topic/thematic structure: **only** with an explicit stability assessment across seeds and preprocessing choices. Learned topics are not treated as ground truth, and if they are unstable that fact is reported rather than the prettiest run.
- Funding structure across CIHR / NSERC / SSHRC / CFI, reported with the SSHRC under-linkage of pilot finding 6 stated alongside.

8. Outputs and release

1. **Dataset** (CSV + Parquet; Frictionless schema): one row per work, carrying route provenance, the five linkage flags, language, genre, tier, classifier score, and audit status.
2. **Validation report**: design-weighted estimates, confusion matrices, and the differential bias report, **including the errors**, not only the headline accuracy.
3. **Code**: the full pipeline, `targets` DAG, Docker image, lockfiles.

4. **Documentation**: Quarto site, a **Datasheet for Datasets**, and this protocol with its registration stamp.
5. **A bilingual explorer** for people who do not write code.

Versioned on GitHub; each release archived to **Zenodo with a DOI**. Public corrections accepted against a transparent version history.

9. Ethics and governance

- The dataset describes **living people**. Individuals are represented only by what they have published.
- **No sensitive identity is inferred or released.** No gender, ethnicity, Indigeneity, or other protected characteristic is imputed from names, affiliations, or text.
- Where Indigenous-governed data or communities are implicated, **no derived labels are released**, and appropriate governance is sought rather than asserted. Invoking OCAP® without an Indigenous partner would be compliance theatre and this study does not do it.
- A public correction mechanism allows individuals to contest or amend records about them.

10. Reflexivity

I am an independent researcher with no institutional affiliation. I am therefore a member of the population this dataset structurally cannot see: affiliation-based country assignment fails on precisely such records, and 64% of the search space carries no affiliation string at all (pilot finding 2). The **CA-FUND** clause in the primary estimand and the **non-retrieved stratum** in the audit exist because of that failure mode.

I am also, on the primary estimand, plausibly *in* the dataset. I will not code any record in the validation sample that includes my own work, and any such record is routed to the second coder and an adjudicator.

11. Deviations

Any departure from this protocol will be recorded in `DEVIATIONS.md` with its date, its reason, and whether it was made before or after the outcome data were seen. Silent revision is itself a research-integrity failure, and this is a metaresearch project.

Appendix A. Screening rubric v3.0: LOCKED, and the instrument the funded work runs under

Status: LOCKED, 2026-07-13. Supersedes v2.2. No screening ran under v3.0, which lived for hours: an adversarial review found that its Murad quotation inserted the word “treatment” into a sentence this document calls verbatim (D29). v3.1 corrects that quotation from the PDF, fixes the Kataoka page range, aligns the funder clause with the protocol, corrects two glosses against their sources, and adds the study-design vocabulary that v3.0 demanded but never listed. Since D29, `pilot/check_quotes.R` fails the build if any quotation marked verbatim is not a substring of its cited PDF.

No number produced so far was made under v3. The pilot ran under v1.0; the 179-work re-screen under v2.0. Both instruments are kept byte for byte, so every figure stays attributable to the text that produced it. v3 is the instrument the **full screen** runs under, and the **v2-to-v3 difference is reported as a finding**.

What changed, and why it is not cosmetic

v1 and v2 had a tier hierarchy: **T1 core / T2 adjacent / T3 contextual**. That structure had a defect at its root, and the data found it before I did.

> ***“Adjacent” is defined by negation.*** *It means* not-T1*. A category whose only content is “near the important one” cannot be applied consistently, and* ***OUT-vs-adjacent was the largest tier confusion at every sample size we measured.*** *The models were not failing.* ***The category was empty.***

It was also a value judgment I had no standing to make. Calling science and technology studies “adjacent” says STS is peripheral to metascience. **STS is a field with its own founding literature, its own journals, and its own account of what studying science means.**

v3 deletes the hierarchy. Every tradition is a **category with its own positive definition, taken from its own literature, quoted verbatim from a source in `docs/reference/definitions/`.** There is no residual bucket. A record may carry **more than one** category, because the traditions genuinely overlap and forcing a single label is what created the problem.

Nothing here is cited from memory. Every quotation below was read from the PDF named beside it. In a project whose worst deviation (D26) was that its own definition of metaresearch was mine and uncited for its entire life, a recalled citation would be the single worst thing this document could contain.

The definitional problem is the field’s, not this project’s, and the field says so in print

- **Puljak L, Lovric Makaric Z, Buljan I, Pieper D.** *What is a meta-epidemiological study? Analysis of published literature indicated heterogeneous study designs and definitions.* J Comp Eff Res. 2020. doi:10.2217/cer-2019-0201 > Analysed **175 information sources**. *“Definitions of meta-epidemiological studies varied and some studies used the term meta-epidemiological study to describe methodological research-on-research studies.”* Conclusion: *“Research community would benefit from consensus about definition of meta-epidemiological study.”*

- **Kataoka Y, et al.** J Clin Epidemiol. 2023;154:219-220. > A **published dispute** with Puljak, opening by naming *“the currently confusing nomenclatures around ‘meta-epidemiology’ and related terms such as ‘meta-research’ and ‘research-on-research’.”*
- **Stevens ER, Laynor G.** *Recognizing the value of meta-research and making it easier to find.* J Med Libr Assoc. 2023;111(4). doi:10.5195/jmla.2023.1758 > *“The breadth of meta-research presents a significant challenge for identifying published meta-research studies,”* and it notes **the lack of a MeSH heading** for meta-research.

Those are claims about humans. The pilot’s numbers are claims about **models disagreeing under one rubric**. They are consistent, they are not the same estimand, and **only the human audit closes the gap** (`DEVIATIONS.md` D27).

The categories

A work may carry **more than one**. **Categories are not ranked and none is primary: the released landscape is the union of the defined categories, with every overlap reported.** The call’s primary estimand is reported over `metaresearch` because that is the field the call names; that is a **reporting choice, not a rank**, and each of the other traditions gets the same per-category estimate.

1. `metaresearch`

> ***“Meta-research is an evolving scientific discipline that aims to evaluate and improve research practices. It includes thematic areas of methods, reporting, reproducibility, evaluation, and incentives (how to do, report, verify, correct, and reward science).”***
>
> *Ioannidis JPA, Fanelli D, Dunne DD, Goodman SN.* Meta-research: Evaluation and Improvement of Research Methods and Practices. *PLoS Biol. 2015;13(10):e1002264. See also Ioannidis JPA. PLoS Biol. 2018;16(3):e2005468; METRICS, Stanford: https://metrics.stanford.edu/research*

Every `metaresearch` record carries a **`domain`**, taken verbatim from those five thematic areas and glossed by Ioannidis’s own parenthetical:

`domain`	Ioannidis’s gloss	operationally
`methods`	*how to **do** science*	design, conduct, analysis, bias, statistical practice, evidence-synthesis methodology; Ioannidis’s own description of this area ends *“research integrity and ethics”*
`reporting`	*how to **report** science*	reporting standards and completeness
`reproducibility`	*how to **verify** science*	replication, data and code sharing, transparency
`evaluation`	*how to **correct** science*	peer review and editorial process, research assessment, metrics

`domain`	Ioannidis's gloss	operationally
`incentives`	*how to **reward** science*	funding allocation, careers and the research workforce, reward structures

RULE 1 stands (carried from v2; it resolved the largest single seam): *a work is `metaresearch` on methods only if its object is methods **as used, reported or performed by researchers**. Developing, deriving, validating or teaching a technique for a domain is OUT, however generic the technique.* **An instance of a research practice is not a study of it.**

2. `metaepidemiology` — CONTESTED IN PRINT. Both definitions are coded.

There is **no consensus definition** (Puljak 2020, above). **This rubric does not invent one.** It codes **both published definitions** and **flags where they disagree**, because the dispute is live, it is between named researchers in peer-reviewed journals, and hiding it inside a single label would destroy the only evidence of it.

- **`metaepi_narrow`** — Murad MH, Wang Z. *Guidelines for reporting meta-epidemiological methodology research.* Evid Based Med. 2017;22(4):139-142. doi:10.1136/ebmed-2017-110713 > *"Meta-epidemiological studies adopt a systematic review or meta-analysis approach to examine the impact of certain characteristics of clinical studies on the observed effect and provide empirical evidence for hypothesised associations." > > (v3.0 quoted this sentence as 'the observed **treatment** effect'. The word is not in the paper; it narrowed Murad's definition while claiming his authority for the narrowing, and it survived a LOCK. D29.)*
- **`metaepi_broad`** — Kataoka Y, et al. J Clin Epidemiol. 2023;154:219-220. *(Not 219-221: page 221 is Puljak's reply, which is the other side of the dispute.) > "The greatest common defining concept of 'meta-epidemiological studies' would be **any studies in which the unit of analysis is a study, not a patient**."*

Where the two differ, the record is flagged, and the size of that class is reported as a finding. A published dispute is a boundary to be *measured*, not settled by fiat.

3. `bibliometrics` (scientometrics)

> ***"Scientometrics is the study of the quantitative aspects of the process of science as a communication system. It is centrally, but not only, concerned with the analysis of citations in the academic literature."***
>
> *Mingers J, Leydesdorff L.* A Review of Theory and Practice in Scientometrics. *Eur J Oper Res. 2015. doi:10.1016/j.ejor.2015.04.002 Founding work: Price DJ de Solla.* Networks of Scientific Papers. *Science. 1965;149(3683):510-515.*

Boundary with `metaresearch`, and it resolves v1's self-contradiction (v1's T1 bullet and its own worked example gave opposite answers): a study that **characterises a literature** by quantitative means is `bibliometrics`. A study that uses bibliometric means to **make a claim about research practice** (citation distortion, reporting trends, citation of retracted work) is **also `metaresearch`**, and carries both. **v3 does not force the choice; that forcing was the bug.**

4. `sts` (science and technology studies)

> ***"co-production is shorthand for the proposition that the ways in which we know and represent the world (both nature and society) are inseparable from the ways in which we choose to live in it." "The idiom of co-production … represents a major synthesis of scholarship in science and technology studies (S&TS)."***
>
> *Jasanoff S (ed.).* States of Knowledge: The Co-Production of Science and the Social Order. *Routledge, 2004.*
>
> ***"We study science in action and not ready made science or technology; to do so, we either arrive before the facts and machines are blackboxed or we follow the controversies that reopen them."***
>
> *Latour B.* Science in Action: How to Follow Scientists and Engineers through Society. *Harvard University Press.*

Operational test: the work studies **science or technology as social practice**: how knowledge and social order are produced together, how facts are made rather than found, how controversies open and close. **The object must be science, research, or technoscientific knowledge**, at any period. *(v1 restricted history and philosophy of science to "contemporary research practice", an unexamined adverb that silently excluded the entire history of Canadian science. It is deleted.)*

5. `scholarly_communication`

> ***"Scholarly communication is a rich and complex sociotechnical system formed over a period of centuries." "…used here in the broader sense to include the formal and informal activities associated with the use and dissemination of information through public and private channels."***
>
> *Borgman CL.* Scholarship in the Digital Age: Information, Infrastructure, and the Internet. *MIT Press, 2007.*

6. `open_science`

> ***"Open science is a set of principles and practices that aim to make scientific research from all fields accessible to everyone for the benefits of scientists and society as a whole." "Open science is about making sure not only that scientific knowledge is accessible but also that the production of that knowledge itself is inclusive, equitable and sustainable."***
>
> *UNESCO.* Recommendation on Open Science *(introduction), 2021.* ***193 countries adopted it.***

The second sentence is the reason this category is not a synonym for "open access": UNESCO's definition makes **inclusivity and equity of knowledge production** part of open science itself, which is the criterion this challenge is judged on.

7. `research_integrity` — defined by its OBJECT; the sources do not define the field, and that is recorded

Honest limitation, stated rather than hidden. The two sources supplied define **what counts as misconduct** and **what publication ethics covers**. **Neither defines "research integrity research" as a discipline** the way Ioannidis defines metaresearch or Jasanoff defines STS. So this category is defined by its **object**, and the gap is on the record.

> ***"…the analysis was limited to behaviours that distort scientific knowledge: fabrication, falsification, 'cooking' of data, etc… Survey questions on plagiarism and other forms of professional misconduct were excluded."***
>
> *Fanelli D.* How Many Scientists Fabricate and Falsify Research? A Systematic Review and Meta-Analysis of Survey Data. *PLoS ONE. 2009;4(5):e5738.*
>
> *Publication ethics, per **COPE** (Committee on Publication Ethics),* Ethics Toolkit for Journal Editors and Publishers.

Operational test: the object is **research misconduct, questionable research practices, retraction, authorship, or publication ethics**, *as an object of study*. **Fanelli's boundary is adopted and its exclusion is carried:** fabrication, falsification and data cooking are in; **plagiarism and professional misconduct are recorded but marked, because Fanelli's own scope excludes them** and a rubric that silently widened his boundary while citing him would be misquoting its own source.

This category overlaps `metaresearch` heavily: Ioannidis's own description of the *methods* area ends *"research integrity and ethics"*, so integrity research is inside metaresearch by his definition. **That overlap is expected and allowed**, because v3 is multi-label. It is not a hierarchy.

`insufficient_payload`

Not a judgment; a statement that the record **cannot be screened** (a scraping artifact as a title, boilerplate in the abstract field, a title and abstract from different papers, spam). **Reported, never silently dropped**: the size of this class is a finding about OpenAlex.

`OUT`

None of the above. **"OUT" now means "in no defined category", not "not near the important one."** That is the whole change.

Canadian linkage

- `CA_AFF` — at least one Canadian institutional affiliation. **Warning, measured:** OpenAlex misassigns these. **At least 17,466** works enter the frame with, as their *only* Canadian institution, one of six suspect assignments: a parse failure that is not an institution at all (`Impact`, attached to a Spanish COVID essay) or a real organization attached to papers it has nothing to do with (`Discovery Air (Canada)` on a Brazilian linguistics paper). That is a **lower bound** from six audited strings (finding 27), and the records are **flagged for audit, not deleted**. A Canadian affiliation is *evidence*, not proof.
- `CA_FUND` — $\geq$1 funder whose OpenAlex `country_code` is Canada: an **external criterion, not a curated list**. *(An earlier four-funder list (CIHR, NSERC, SSHRC, CFI) failed twice: three of its four*

hardcoded IDs pointed at other funders, and even corrected it missed Canada Research Chairs, the Government of Canada, NRC, Mitacs, Genome Canada, and every provincial agency. OpenAlex knows 2,182 Canadian funders; `R/pin_entities.R` pins them.)

- **`ABOUT_CA_SYSTEM`** — the Canadian **research system or academy** is a substantive object. *A Canadian institution as setting is not this.*
- **`ABOUT_CA_TOPIC`** — Canada is a substantive subject, whether or not research is the object.

(v1 had one boolean and it could not separate "Canadian data, universal claim" from "a claim about Canada". Four screening agents proposed this split independently.)

Study design

Coded for every work, from **this vocabulary and no other** (v3.0 said "coded for every work" and never listed the values, which is D20's exact shape: a field every screener codes and no two code alike):

`study_design`	MEDLINE publication type it is validated against
`randomized_trial`	Randomized Controlled Trial
`nonrandomized_trial`	Controlled Clinical Trial
`observational`	Observational Study
`systematic_review`	Systematic Review
`meta_analysis`	Meta-Analysis
`case_report`	Case Reports
`qualitative`	*(none; ships as a score, marked unvalidated)*
`simulation_or_modeling`	*(none; ships as a score, marked unvalidated)*
`bench_or_experimental`	*(none; ships as a score, marked unvalidated)*
`theoretical_or_conceptual`	*(none; ships as a score, marked unvalidated)*
`not_applicable`	*(the work is not empirical research: editorials, datasets, software)*
`design_other`	*(none; ships as a score, marked unvalidated)*

Validated against MEDLINE publication types, per record and not per year. *(v3.0 called these "human-indexed by NLM", which is false; v3.1's first correction said "human-indexed until April 2022", which is also false. NLM stopped fully human indexing as universal practice around 2011 and completed automation in 2022, so a YEAR CUTOFF cannot separate the eras. The reference subset is defined by NLM's **record-level `IndexingMethod`** field: absent = fully human, `Curated` = automated with human review, `Automated` = machine. Human records are the reference; the others measure **agreement with another machine** and are reported as such.)*

Class-specific calendar eligibility, because a publication type that did not exist cannot be a negative. *Observational Study* entered the vocabulary in **2014** and *Systematic Review* in **2019**. A 2008 cohort study carries no *Observational Study* tag, and treating that absence as "not observational" would train and score the model against a censoring artifact. Each design class is therefore validated **only on records published after its type existed**, and that eligibility window is prespecified per class. Where MEDLINE does not reach (most of the physical sciences, humanities and social sciences), the label ships as a **score, not an assertion**, and is **marked unvalidated**. *The dataset states, per field, which of its own annotations it has earned the right to assert.*

Genre

One vocabulary, defined here, and `screening-schema-v3.json` QUOTES this list rather than paraphrasing it.

`empirical · review · methods · commentary · editorial · protocol · dataset · software · other`

(In v1 the rubric and the schema named ***two different vocabularies for this field****, overlapping on two values out of thirteen. Every screener was handed both and obeyed both privately; the validator checked `tier` and had never been asked to look at `genre`, so 16,800 labels passed every check that ran. See D20 and finding 24.* ***A codebook is code. It gets a test.****)*

`tier` is RETIRED

v3 replaces it with `categories`, which is **multi-label**. The values `T1`, `T2`, `T3` and `OUT` remain in the **legacy** `screening-schema.json` **only so that the v1 and v2 label files stay readable**; the v3 contract, `screening-schema-v3.json`, **does not contain the field**, so no new screening can emit them. `OUT` in v3 means an empty `categories` array: *in no defined category*, not *near the important one*.

Confidence

`high` / `medium` / `low`. **If the abstract is missing, judge on the title alone and set confidence to `low` unless the title is unambiguous.** The schema **quotes** this sentence; it does not paraphrase it (D24).

`low` is a legitimate answer. A screener that never says `low` is not being careful; it is being confident.

Sources

Every quotation above was read from a PDF in **`docs/reference/definitions/`**, which is committed. Nothing is cited from memory, and since D29 that is a checked property, not a promise: `pilot/check_quotes.R` fails the build if any quotation marked verbatim is not a substring of its cited PDF. `pilot/check_instrument.R` fails the build if this rubric and `screening-schema-v3.json` disagree about any field, in either direction.

Appendix B. Screening rubric v1.0: SUPERSEDED, and the instrument every pilot number was produced under

Locked before screening. Version-controlled. Any change is a new version with a dated entry in `DEVIATIONS.md`.

Both machine screeners and both human coders work from this document and nothing else. The rubric is the instrument; if two coders disagree, the rubric is what they are adjudicated against.

The unit

One work (journal article, preprint, book chapter, report, dataset, or software), identified by its OpenAlex ID. You see: title, abstract, publication year, language, venue, OpenAlex topic and field, Canadian institutional affiliations, funders.

The question

Is the object of study *research itself*?

Metaresearch studies how research is done, reported, funded, evaluated, disseminated, or governed. The test is what the work is *about*, not what field it appears in and not what vocabulary it uses.

A cardiology trial is not metaresearch. A study of *how cardiology trials report their outcomes* is.

Tier

Assign exactly one.

T1: Core metaresearch

The primary object of study is research itself. Typical objects:

- research **methods** and their properties (bias, reporting, statistical practice, study design)
- **reproducibility**, replication, data and code sharing
- **research integrity**: misconduct, retraction, questionable research practices, authorship
- **peer review** and editorial process *as objects of study*
- **research evaluation**, metrics, rankings, funding allocation
- **scholarly communication**: publishing models, open access, preprints *as objects of study*
- **bibliometrics and scientometrics** applied to a research literature
- the **research workforce**: careers, equity, training, capacity
- **evidence synthesis methodology** (not an individual systematic review's clinical answer, but work *about* how syntheses are done)

T2: Adjacent

Empirically or conceptually grounded work on science as a social system that is not framed as metaresearch but studies it:

- **science and technology studies (STS)**: social studies of scientific knowledge, practice, controversy
- **library and information science**: information behaviour of researchers, repositories, discovery
- **research infrastructure** studies: platforms, data repositories, identifiers, standards
- **history and philosophy of science** *when the object is contemporary research practice*
- **research policy** analysis grounded in evidence

T2 is where the field's diversity lives. Be generous here: a paper does not have to call itself metaresearch to be studying research.

T3: Contextual

Relevant to the ecosystem, mapped and released, but **never pooled into the analytic corpus**:

- editorials, commentaries, opinion pieces about research practice
- policy documents and position statements
- announcements of infrastructure, tools, or initiatives
- news, obituaries, corrections, letters

OUT: Not metaresearch

Everything else. Includes, and this is the most common error:

- **A study that uses a method is not a study of the method.** A meta-analysis answering a clinical question is OUT. A study of how meta-analyses handle heterogeneity is T1.
- **Mentioning a metaresearch term is not studying it.** A lab paper reporting "assay reproducibility", a methods section saying "peer reviewed", a licence note saying "open access": all OUT.
- **Domain research that happens to be bibliometric in method** but whose object is the domain, not the research: e.g. a bibliometric map of diabetes treatments, produced to summarise diabetes knowledge. Borderline; see below.

The three errors that matter most

These are where the pilot showed retrieval and screening break down. Read them twice.

1. **Polysemy.** `reproducibility` in a biochemistry paper means the assay gives the same reading twice. In a metaresearch paper it means the reproducibility crisis. Only the second is in scope. The same applies to `validation`, `replication`, `bias`, `peer review`, `open access`. **Judge the sense, not the token.**
2. **Method vs. object.** Using bibliometrics $\neq$ studying research. Ask: *if this paper is right, what do we now know more about: a scientific question, or the practice of science?* If the former, OUT.
3. **Absence of vocabulary is not absence of content.** An STS paper about laboratory practice, or an LIS paper about how researchers find data, may use none of the field's keywords and still be squarely T2. **Do not require the vocabulary.** This is the error that makes the field look smaller and more anglophone than it is.

Canadian linkage

Record which apply (this is metadata, not an inclusion test; the estimand is defined separately in `PROTOCOL.md`):

- `CA_AFF`: at least one author affiliated with a Canadian institution
- `CA_FUND`: funded by CIHR, NSERC, SSHRC, or CFI
- `ABOUT_CA`: the Canadian research system is a substantive object of study (not a passing mention, not one country among forty in a global comparison)

Genre

`empirical` · `conceptual` · `editorial/commentary` · `policy` · `infrastructure/announcement` · `other`

OpenAlex's `type` is unreliable and this field is commentary-heavy, so genre is coded, not inherited.

Language of the work

As published. Do not infer from the venue.

Confidence

- `high`: the rubric decides this cleanly.
- `medium`: the rubric decides, but a reasonable coder could differ.
- `low`: genuinely on the boundary, or the abstract is missing/too thin to judge.

`low` is a legitimate answer and is not a failure. Records coded `low` by either screener are routed to adjudication. A screener that never says `low` is not being careful; it is being confident.

Worked examples

Title (abridged)	Tier	Why
"Data sharing practices in 500 randomised trials"	**T1**	The object is reporting practice.
"Reproducibility of a Western blot protocol"	**OUT**	Assay reproducibility. Polysemy trap.
"Trends in citation of retracted papers"	**T1**	Object is the literature's own behaviour.
"Laboratory life in a Canadian genomics centre: an ethnography"	**T2**	STS. No metaresearch vocabulary; still studies research.
"How do researchers discover datasets? A survey"	**T2**	LIS, information behaviour of researchers.

Title (abridged)	Tier	Why
"Effectiveness of statins: a network meta-analysis"	**OUT**	Uses a synthesis method to answer a clinical question.
"How network meta-analyses handle inconsistency: a review of 200 reviews"	**T1**	Object is the method.
"Announcing the Canadian Research Data Repository"	**T3**	Infrastructure announcement. Mapped, not pooled.
"Open access mandates are failing early-career researchers" (editorial)	**T3**	Commentary about research practice.
"Bibliometric analysis of diabetes research 2000–2020"	**T2**	Borderline. Bibliometric method, and the object *is* a research literature, so it studies research output. Code T2, `medium`.
"Le libre accès dans les revues québécoises : état des lieux"	**T1**	Scholarly communication, object is publishing practice. French.

Instructions to the screener

Read the title and abstract. Apply the rubric. Do not use outside knowledge of the authors or the journal's reputation. If the abstract is missing, judge on the title alone and set confidence to `low` unless the title is unambiguous.

Return the schema in `docs/protocol/screening-schema.json`. One object per work. No prose, no preamble.

Appendix C. The seam census: how v2 was derived, published before it was resolved

Status: PROPOSED. Not locked, not applied. The v1 rubric (`docs/protocol/rubric.md`) is the instrument the pilot ran under, and changing it after seeing the data would be fitting the instrument to the data. This document is the **evidence for a revision**, to be locked *before* the full screen and recorded in `DEVIATIONS.md` as a versioned change.

Where this evidence comes from, and why it is unusual

Fifteen Opus screening agents classified 5,000 works from the real frame against the locked v1 rubric, in two cohorts (five agents on works 1 to 2,000; ten agents on works 2,001 to 5,000). They worked **independently**, on **disjoint chunks**, with **no knowledge of each other**, and were asked, at the end, which boundary cases recurred.

They converged. Not loosely: agents seeing entirely different works independently reported the *same* underspecified distinctions, often naming the same axis in the same terms, and the second cohort reproduced the first cohort's seams without having seen them. That convergence is what makes this evidence rather than opinion. A distinction that fifteen frontier-model screeners, reading one rubric, cannot apply consistently **is not a distinction. It is a wish**, and the disagreement it produces is not noise to be averaged away: it is the field's boundary showing you where it actually is.

Finding 22 measured the consequence at the tier level. Finding 24 measured something worse: **the instrument contradicts itself**, and the screeners found that too.

Two facts about v1 that are not opinions

Before the seams, two defects that can be checked against the text rather than argued about.

A. The rubric and the schema name different vocabularies for `genre`. `docs/protocol/rubric.md` says `empirical · conceptual · editorial/commentary · policy · infrastructure/announcement · other`. `docs/protocol/screening-schema.json` says `empirical · review · methods · commentary · editorial · protocol · dataset · software · other`. **They overlap on two values out of thirteen.** Every screener was handed both and told to obey both; each invented a private reconciliation, and they did not invent the same one. Four separate agents reported this unprompted. See finding 24 and `DEVIATIONS.md` D20. It is now quarantined in `docs/protocol/known-defects.json` and `make lint` fails on any *further* contradiction between the two documents.

B. Two of the six verbs in the rubric's own definition have no tier bullet. The definition says metaresearch studies how research is *"done, reported, funded, evaluated, disseminated, or governed."* Four of those verbs map onto T1 bullets. Two do not:

- **disseminated**: `scholarly communication` covers publishing models, open access and preprints, which is dissemination *of outputs into the literature*. Nothing covers **knowledge translation and implementation science**, i.e. how research reaches practice. Four agents flagged it, and all four coded such work OUT while saying they were unsure.

- **governed**: nothing in T1 covers **research ethics, human-subjects governance, or research-governance institutions**, though `governed` is the promise the definition makes.

A definition that names six objects and operationalizes four is not a strict instrument. It is an instrument with two undefended doors.

The seams, ranked by how many independent screeners raised them

1. Domain methodology vs. research methods, and its sharper form: ***developing*** a method vs. ***studying*** one (raised by 15 of 15)

The single largest source of disagreement, named by every screener in both cohorts.

T1 says "research methods and their properties (bias, reporting, statistical practice, study design)". Read one way that covers every methods paper in every field: a SNP-caller benchmark, a neuroimaging pipeline, a photoionization model, a thyroid-nodule segmentation network. Read another way it covers only cross-cutting research practice.

The second cohort sharpened it into a distinction the first cohort had not isolated. **The bullet never says whether *proposing a new method* counts, or only *studying methods as they are used across a literature*.** A new high-breakdown multivariate estimator in the *Canadian Journal of Statistics* is T1 by the letter and OUT by the spirit; every worked example in the rubric is a study *of a literature*, so nothing anchors the line. Screeners independently converged on the same rule and independently said they could not defend it from the text:

> *"results depend on the analytic choice" studies → T1; building a tool for a domain task → OUT.*

A paper showing feature-selection bias inflates published classification accuracy is T1. A paper applying a new model to patient satisfaction is OUT. But a paper showing that a standard oceanographic estimator underestimates respiration threefold *studies a method's properties across a literature* and its payoff is a domain claim, and nothing in v1 decides it.

> ***Proposed rule.*** *T1 covers work whose object is* research practice across a literature *(reporting, bias, synthesis methodology, statistical practice, analytic-choice sensitivity),* ***including method-comparison and method-critique studies whose finding generalizes to how research is done****. It does* **not** *cover development of a technique for a domain task,* ***nor the introduction of a new method as such****, unless the work's contribution is a claim about research practice. The discriminator, stated as the rubric's error 2 already almost states it:* if this paper is right, what do we now know more about: a scientific question, or the practice of science? *Add both worked examples, and add a new-method example explicitly.*

2. Measurement and psychometrics (15 of 15)

Instrument-validation studies recur constantly: a MoCA norms paper, a TAS-20 refinement, a French-Quebec CCC-2 adaptation, a systematic review of minimal important differences for patient-reported outcomes, ActiGraph step-count validity, ICD-10 code validation, external validation of six published prediction equations.

Is validating a measurement instrument "research methods and their properties" (T1), or is it domain research whose object is a clinical construct (OUT)? The rubric flags `validation` as a polysemy trap *without saying how to resolve it*, which is the worst of both worlds: it tells the screener the word is dangerous and then leaves the danger.

Every screener invented a line, and **they invented compatible but unstated lines**: single-instrument validation in a domain → OUT; methodological work about *how instruments get built or should be chosen* (COSMIN-style, core outcome sets, cross-instrument reviews) → T1. One agent noted the hardest case: a validity review of a device *explicitly justified by its use as a research reference standard*.

> ***Proposed rule.*** *Validation of a* specific instrument for a domain construct *is* ***OUT****, even when the instrument is used in research. Work whose object is* instrument-development methodology*,* cross-instrument comparison*,* core outcome set methodology*, or* how a field chooses what to measure *is* ***T1****. State it, with both worked examples, and resolve* `validation` *explicitly in the polysemy section.*

3. T3's document-type bullets are unscoped (13 of 15)

T3 lists "news, obituaries, corrections, letters" and "announcements of infrastructure, tools, or initiatives" **without saying the item must be about research**. Only the *editorials* bullet carries the qualifier "about research practice". Taken literally, T3 sweeps in every clinical case letter, every domain erratum, and every bioinformatics software release in existence.

Screeners split, and said so: a reviewer-acknowledgement list is T3 (its object *is* peer review); an erratum on a binder-jet metallurgy paper is OUT (its object is metallurgy). Under the literal reading both are T3, and T3 fills with domain corrections.

> ***Proposed rule. Every T3 bullet requires the item's object to be research or the research ecosystem****, not merely its genre. A letter about a domain result is OUT. A letter about peer review is T3. T3 infrastructure means* ***research-ecosystem*** *infrastructure: domain analysis software and scientific instruments are OUT. Move the "about research practice" qualifier from the editorials bullet up into T3's heading, where it governs all four.*

4. Peer review *performed* is not peer review *studied* (5 of 15, and entirely new in cohort 2)

T1 covers peer review "as objects of study". The frame turns out to contain peer review **as an artifact**: published referee reports, open peer-review records, PREreviews of preprints, "Discussion of" comments. OpenAlex even has a `peer-review` record type.

These are *instances* of the review system whose content is dogwhelks, or hydraulic modeling, or a bacteriology preprint. The rubric never says whether the review system's own outputs are in the map. Screeners split them from reviewer-acknowledgement lists on instinct.

> ***Proposed rule.*** *A peer-review artifact whose* content *is a domain paper is* ***OUT****; it is an instance, not a study, and this is the same rule as error 1 (a study that uses a method is not a study of the method). A peer-review artifact whose object is the review process itself is* ***T3****. Say so, because the frame is full of them and they are currently coded by coin-flip.*

5. History of science falls into a hole the rubric dug (7 of 15)

T2 admits history and philosophy of science **only "when the object is contemporary research practice"**. The STS bullet immediately above it carries **no such qualifier**. So a historical social study of scientific knowledge satisfies one bullet and fails the other, and the screeners split on exactly that.

The consequence is not a rounding error. Coded strictly, **the history of Canadian science largely drops out of the map**: the politicization of Ayurveda around Indian independence, how cybernetics spread through Latin America, how the Université de Montréal built its psychiatry training from a 1962 survey of US programs, Penfield and Jasper, a history of the Canadian archival profession. For a project whose inclusiveness criterion is about *traditions*, silently excluding an entire tradition on an unexamined adverb is the worst available outcome.

> ***Proposed rule. Delete "contemporary".*** *Qualify the STS and HPS bullets identically: the object must be* ***science, research, or the research ecosystem****, at any period. If a date restriction is wanted it belongs in the estimand, not smuggled into one tier bullet where it silently deletes a tradition.*

6. LIS and STS are scoped asymmetrically (6 of 15)

T2's LIS bullet says "information behaviour **of researchers**". The STS bullet carries **no equivalent qualifier**, which makes STS-styled work on non-science technology (network neutrality, Sidewalk Labs' urban socio-technical imaginaries) genuinely undecidable, and lets general library-service evaluation drift in under "be generous".

> ***Proposed rule.*** *Qualify both bullets the same way: the object must be* ***science, research, or the research ecosystem****. This is the seam that produces the OUT-vs-T2 confusion finding 22 measured, so it matters most for the inclusiveness criterion.*

7. Bibliometrics: T1 and the worked example flatly contradict each other (6 of 15)

T1 lists "bibliometrics and scientometrics applied to a research literature". The worked example sends "Bibliometric analysis of diabetes research 2000-2020" to **T2**, with the reasoning *"the object is a research literature"*. **That is the T1 bullet's own criterion, used to justify T2.** Both cannot be right, and screeners followed different ones.

> ***Proposed rule.*** *Pick one and delete the other. Recommended: a bibliometric/scoping study that* characterizes a literature *(who publishes, where the gaps are) is* ***T2****; one that* studies research practice through bibliometric means *(citation distortion, reporting trends, retraction citation) is* ***T1****. Then fix the worked example so it stops contradicting the bullet.*

8. The research workforce: the rubric says "research", the screeners still could not apply it (8 of 15)

Correction to an earlier draft of this document, which claimed the rubric fails to say "research workforce". It says it. The bullet reads "the **research workforce**: careers, equity, training, capacity". The seam is real but it is not the one I first wrote down, and it is worth being precise about, because misdescribing your own instrument is how you come to fix the wrong thing.

The actual failure is at the **boundary of the role**, not the wording:

- Academics whose role spans **teaching and research** have no home: bias in student evaluation of teaching, faculty development for clinician teachers, a protocol for developing nursing academic leaders who lead both.
- **Higher-education studies** generally: participation policy, university writing centres, spending and student engagement across 18 Ontario universities.
- **Professional and clinical training**: nursing retention, medical-student debt, surgical curricula, interprofessional education. These are professional populations, not the research workforce, and screeners flagged them as the most common near-miss after the polysemy trap.

> ***Proposed rule.*** *Keep "research workforce" and add the exclusion explicitly:* the professional (non-research) workforce, and classroom or clinical pedagogy, are OUT, including when the population is academics acting in a teaching role. *Add a worked example on each side of that line, because it is a line, not a wish, and it can be drawn.*

9. Records with no judgeable content (6 of 15)

The frame carries bare-DOI titles, `.mat` filenames, "Additional file 3 of…", GBIF occurrence downloads, Crossref persistence boilerplate (which is *literally text about persistent identifiers*, a scholarly-infrastructure keyword trap), DOAJ open-access boilerplate in the abstract field (a false metaresearch signal), an SEO spam page, a hockey podcast, a woodworking magazine. One agent found a record whose **title and abstract are about different papers** (a body-image title with a bacterial-injectisome abstract).

These are all coded OUT/low today, which **inflates the low-confidence rate and pollutes the adjudication queue** with records that have nothing to adjudicate.

> ***Proposed rule.*** *Add an* ***`insufficient_payload`*** *flag, distinct from OUT. It is not a screening judgment; it is a statement that the record cannot be screened. It must be* reported*, not silently dropped: the size of this class is a finding about OpenAlex, and the audit samples it.*

10. Knowledge translation and research governance are promised and never delivered (5 of 15)

See fact **B** above. The definition says research is "disseminated" and "governed"; no tier bullet covers knowledge translation, implementation science, or research ethics and governance. Screeners consistently coded implementation work OUT while saying the preamble seemed to want it in.

> ***Proposed rule.*** *Decide, in the text, and say which way. Recommended:* ***knowledge translation and implementation science, where the object is the uptake of research findings, is T2****;* health-service delivery that happens to involve an intervention is OUT. ***Research ethics and governance as an object of study is T1.*** *Either add the bullets or remove the verbs from the definition, but do not keep promising six things and operationalizing four.*

11. Guidelines, consensus standards, and reporting standards (4 of 15)

The COBIDAS neuroimaging best-practice report, a Delphi harmonizing surgical definitions *explicitly so results can be compared across centres*, a core-outcome-set consensus statement. These are **reporting-standard work by motivation** and **position statements by form**, so T1's methods bullet and T3's "position statements" bullet both claim them.

> ***Proposed rule.*** *A consensus standard whose object is **how research should be reported, measured, or compared** is **T1**, regardless of its genre being a position statement. A position statement about policy is T3. Genre does not decide tier; the object does. (This is the same principle as seam 3 and it should be stated once, globally.)*

12. Supplements should inherit their parent (3 of 15)

"Additional file N of *[a scoping review of airway registries]*" has no content of its own. Screeners judged the implied parent, which is sensible and is **nowhere in the rubric**.

> ***Proposed rule.*** *State it: a supplement inherits its parent's tier, or is flagged `insufficient_payload` when the parent cannot be identified.*

13. `about_ca` is near-degenerate, and it carries the secondary estimand (4 of 15, and measured)

`about_ca` is defined as "the Canadian research system is a **substantive object of study**". Three problems, and the third is measured, not argued.

- T3 items are **not studies**, so the definition does not literally apply to them, and screeners diverged on an NRC audit plan and a *Canadian Historical Review* editorial.
- **A Canadian institution as *setting* is not `about_ca`.** A Quebec COVID cohort is not about the Canadian research system; an analysis of Canadian law faculties' tenure norms is. Screeners applied this correctly but it is nowhere written.
- **It fires on about 1% of the sample.** In the `about_only` stratum, whose *entire retrieval basis* is that Canada appears in the title or abstract, screeners marked `about_ca` true for **1.8% (Opus), 1.3% (GPT-5.6), 0.5% (Grok)**. The route retrieves *"Canada is mentioned"*; the rubric asks for *"the Canadian research system is studied"*. These are different questions, and the gap between them is the polysemy finding again, applied to the country term.

The consequence is structural: **the secondary estimand of this study ("metaresearch about the Canadian research system") currently rests on a few dozen sampled events, and the three models disagree about it by more than they disagree about tier.**

> ***Proposed rule.*** *Extend the definition to non-study genres ("substantive object", not "object of study"); add the setting-vs-object exclusion with both worked examples. And, in the* protocol *rather than the rubric: the audit must **oversample `about_ca` candidates**, because at a 1% hit rate the secondary estimand cannot be estimated from a sample designed for the primary one.*

Two further polysemy traps, found in the wild

v1 names `reproducibility`, `peer review`, `open access`, `open science`, `validation`, `replication`, `bias`. The screeners hit more, and hit the named ones in unnamed senses:

- **`validation / evaluation`**: institutional policies on *student* assessment, clinical *instrument* validation, health-professions competence assessment. Named as a trap, never resolved (seam 2).
- **`audit`**: financial audit, audit fees, clinical audit.

- Plus the observed traps: `"peer reviewed: yes"` as a repository artifact; judicial *standard of review*; prison *misconduct*; managerial *reporting*; DOAJ and Crossref boilerplate that is *literally about scholarly infrastructure*; and, memorably, **“Canada dry” as a French idiom**.

What this does NOT license

This document does not change the pilot’s numbers, and must not. Every figure in the proposal was produced under v1, and re-screening under v2 to get a better number would be exactly the failure this project exists to object to.

The correct sequence, and the one the protocol commits to:

1. Lock v2 **before** the full screen, with this evidence attached.
2. Record the change in `DEVIATIONS.md` as a versioned rubric revision, and **delete the quarantine entry** in `docs/protocol/known-defects.json` so `make lint` enforces instrument self-consistency from that point on.
3. Re-screen the 5,000-work sample under v2 and **report the difference against v1 as a finding**: *what did specifying these thirteen seams actually change?* That is a measurement nobody normally makes, and it is available only because the ambiguity was documented before it was resolved.